

JEE MAIN REVISION MASTER: CHEMICAL BONDING & MOLECULAR STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 214 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	VSEPR Theory & Molecular Geometry	109 PYQs	Molecular Shape
Very High	Molecular Orbital Theory (MOT) & Bond Order	72 PYQs	MOT
Very High	Dipole Moment, Formal Charge & Hydrogen Bonding	37 PYQs	Intermolecular

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
VSEPR Theory & Molecular Geometry	$\text{Steric Number } H = \frac{1}{2}(V + M - C + A)$, $\text{LP-LP repulsion} > \text{LP-BP} > \text{BP-BP}$
Molecular Orbital Theory (MOT) & Bond Order	$\text{Bond Order} = \frac{N_b - N_a}{2}$, $\text{Stability} \propto \text{Bond Order}$, $\text{Bond Length} \propto \frac{1}{\text{Bond Order}}$
Dipole Moment, Formal Charge & Hydrogen Bonding	$\mu = q \times d$, $\text{Formal Charge} = V - L - \frac{1}{2}S$, $\mu(\text{NH}_3) > \mu(\text{NF}_3)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 VSEPR Theory & Molecular Geometry (Very High Yield — 109 PYQs)

Core Principle: Prediction of molecular shapes and bond angles using steric number and lone pair repulsions.

■ **Caution / JEE Pitfall:** XeF_4 has sp^3d^2 hybridization with 2 lone pairs; its geometry is octahedral, but its molecular SHAPE is SQUARE PLANAR.

■ **Speed Trick:** In trigonal bipyramidal geometry (sp^3d , e.g. PCl_5), lone pairs ALWAYS occupy equatorial positions to minimize 90° repulsions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Bromine trifluoride autoionizes to form BrF_2^+ and BrF_4^- . The shapes of the cation and anion are respectively ____ and..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "The covalent radii of atoms A and B are r_A and r_B , respectively. The covalent bond length and total length of AB molecule are..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II according to shape. List - I List - II A. XeO_3 B. XeF_2 C. XeO_2F_2 D. XeOF_4 I. BrF_5 II. NH_3 III. ..." [Ans: N/A]

3.2 Molecular Orbital Theory (MOT) & Bond Order (Very High Yield — 72 PYQs)

Core Principle: LCAO principle, molecular orbital filling order for ≤ 14 vs > 14 electrons, bond order calculation, and magnetic character.

■ **Caution / JEE Pitfall:** O_2 has bond order 2 and is PARAMAGNETIC because it has 2 unpaired electrons in degenerate antibonding $\pi^* 2p_x$ and $\pi^* 2p_y$ orbitals.

■ **Speed Trick:** For species with ≤ 14 electrons (B_2 , C_2 , N_2): $\pi 2p_x = \pi 2p_y$ is filled BEFORE $\sigma 2p_z$ due to s-p mixing.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Which of the following pictorial diagram most correctly represents the π^* (π antibonding) molecular orbital between two atoms if..." [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "Given below are two statements : Statement I : $\text{F}_2\text{O} < \text{H}_2\text{O} < \text{Cl}_2\text{O}$ is the correct trend in terms of bond angle. Statement II : $\text{SiF}_4, \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "Pair of species among the following having same bond order as well as paramagnetic character will be..." [Ans: N/A]

3.3 Dipole Moment, Formal Charge & Hydrogen Bonding (Very High Yield — 37 PYQs)

Core Principle: Vector addition of bond dipoles, percentage ionic character, and influence of H-bonding on boiling points.

■ **Caution / JEE Pitfall:** Dipole moment of NH_3 is HIGHER than NF_3 because lone pair and N-H bond moments reinforce each other in NH_3 , but oppose in NF_3 .

■ **Speed Trick:** Intermolecular H-bonding increases boiling point and water solubility (e.g. p-nitrophenol), while intramolecular H-bonding decreases boiling point (o-nitrophenol).

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are..." [Ans: N/A]
- [2026 Morning Shift] Q#4 (MCQ): "Given below are statements about some molecules/ions. Identify the CORRECT statements. A. The dipole moment value of NF_3 is higher than..." [Ans: N/A]
- [2026 Evening Shift] Q#4 (MCQ): "Among H_2S , H_2O , NF_3 , NH_3 and CHCl_3 , identify the molecule (X) with lowest dipole moment value. The number of lone pairs o..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Bromine trifluoride autoionizes to form BrF_2^+ and BrF_4^- . The shapes of the cation and anion are respectively _____ and _____.

- (A) bent, square planar **[Correct]**
- (B) linear, square planar
- (C) bent, see-saw
- (D) linear, tetrahedral

Correct Answer: A

Solution Analysis: First, let us recall the autoionization of bromine trifluoride: $2\text{BrF}_3 \rightleftharpoons \text{BrF}_2^+ + \text{BrF}_4^-$. We have to determine the shapes of the cation and the anion. Central atom: Br Valence electrons of Br = 7 Each F contributes one bond $\rightarrow 2 \text{ Br-F bonds} = 2 \text{ pairs of bonding electrons}$ So, total electrons around Br for shape determination: $7 - 2(\text{by bonding}) - 1(\text{positive charge}) = 4$ These 4 electrons form 2 lone pairs on Br. Hence, the total number of electron pairs around Br = 2 (bonding) + 2 (lone pairs) = 4 electron domains. According to VSEPR theory: 4 electron pairs $\rightarrow$

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#3)

The correct increasing order of C-H(A), C-O(B), C=O(C) and C $\equiv$ N(D) bonds in terms of covalent bond length is :

- (A) $< C < B < A$
- (B) $< C < A < B$
- (C) $< B < C < D$
- (D) $< D < C < B$ **[Correct]**

Correct Answer: D

Solution Analysis: Bond length is inversely related to bond order (as per NCERT idea): triple bond $<$ double bond $<$ single bond (shorter to longer). Also, for single bonds, bond length increases when the bonded atom is larger (so C-O is longer than C-H). Now compare: C-H (A): single bond, but H is very small $\Rightarrow$ short C $\equiv$ N (D): triple bond $\Rightarrow$ very short C=O (C): double bond $\Rightarrow$ short, but longer than triple C-O (B): single bond with larger O $\Rightarrow$ longest So increasing order of bond length is: $A < D < C < B$ Correct option: D

Benchmark Problem 3: JEE Main 2024 (Online) 1st February Morning Shift (Q#4)

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : PH_3 has lower boiling point than NH_3 . Reason (R) : In liquid state NH_3 molecules are associated through Vander Waal's forces, but PH_3 molecules are associated through hydrogen bonding. In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (B) (A) is not correct but (R) is correct
- (C) (A) is correct but (R) is not correct **[Correct]**
- (D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Correct Answer: C

Solution Analysis: The correct answer is Option C: (A) is correct but (R) is not correct. Assertion (A) : PH_3 has lower boiling point than NH_3 . This assertion is true. The boiling point of ammonia (NH_3) is higher than that of phosphine (PH_3). Ammonia has a boiling point of about -33.34°C , whereas phosphine has a boiling point of about -87.7°C . The reason for the higher boiling point of ammonia as compared to phosphine lies in the strength and type of intermolecular forces present in the substances. Reason (R) : In liquid state NH_3 molecules are associated through Vander Waals' force